

Package ‘YuleInfer’

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Title Inference for correlated binary data using Yule's Colligation Coefficient

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Description Estimates and tests a common marginal risk difference for two groups with paired and unpaired binary observations in one or more strata. Accepts subject-level data and aggregated counts. Within-subject association is represented by a Yule, Dallal, Rosner or Donner parameter, shared between treatment groups within each stratum. Provides constrained likelihood estimation, likelihood-ratio, efficient-score and Wald tests, confidence-set inversion, simulation and analytic local power. Includes a unified analysis report, explicit treatment ordering, and numerical convergence and boundary diagnostics.

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YuleInfer-package	<i>Inference for correlated binary data using Yule's Colligation Coefficient</i>
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Description

Likelihood inference for a common marginal risk difference in stratified binary data containing bilateral and unilateral subjects. The two treatment groups share a Yule, Dallal, Rosner or Donner native association parameter within each stratum; parameters may differ across strata.

Details

Start with `yule_import` for one row per subject, or provide aggregate counts to `yule_analyze` for a fit, test, confidence set and readable report in one call. A single stratum and unilateral-only data are supported. `paired_trial` is a synthetic worked example; `otitis_media` contains published age-stratified clinical counts.

The model-neutral `strat_*` functions cover data validation, bilateral probabilities, simulation, likelihood fitting, testing, confidence-set inversion and own-model analytic local power. Existing StratYule functions remain available under the new package name. The likelihood engine includes registered C routines. Building from source requires R's C compilation tools; installing a supplied platform binary does not. Runtime dependencies are limited to R and its standard **stats** and **utils** packages.

For a bilateral subject, K is the number of positive responses, with probabilities $p_0 = 1 - 2\pi + q$, $p_1 = 2(\pi - q)$, and $p_2 = q$. Yule's colligation coefficient is

$$\Upsilon = \frac{\sqrt{q(1 - 2\pi + q)} - (\pi - q)}{\sqrt{q(1 - 2\pi + q)} + (\pi - q)}.$$

This coefficient is also called Yule's Y ; it is not Yule's Q . The classical native coordinates are Dallal's conditional probability q/π , Rosner's conditional-to-marginal risk ratio q/π^2 , and Donner's correlation $(q - \pi^2)/\{\pi(1 - \pi)\}$. See `strat_models` for full feasibility bounds and the scope of these adapted common-risk-difference likelihood procedures. All supported regular models have interior margins and strictly positive bilateral cells. The common effect is the second group's margin minus the reference group's margin. The same per-unit marginal response probability is assumed for unilateral and bilateral subjects in a stratum and group. Subjects are independent and strata are independent.

Common-effect homogeneity and sharing the selected native association are model assumptions. The procedures do not test homogeneity across strata and do not provide robust inference for arbitrary association misspecification. Tests and confidence sets use large-sample approximations. Near-boundary, nonconverged, or singular fits are identified in diagnostics and excluded from regular chi-square inference.

Two registered practitioner vignettes provide complete examples. The first follows a clinical example and subject-level import. The second covers association models, diagnostics, simulation and local power.

```
vignette("getting-started", package = "YuleInfer")
vignette("models-and-inference", package = "YuleInfer")
browseVignettes("YuleInfer")
```

The retained HTML guide links to the vignettes. Use `citation("YuleInfer")` for the software citation.

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See Also

- Prepare and analyse** [yule_import](#), [strat_data](#), [yule_analyze](#).
- Fit and infer** [strat_fit](#), [strat_test](#), [strat_confint](#).
- Generate and plan** [strat_probabilities](#), [strat_simulate](#), [strat_power](#).
- Model assumptions** [strat_models](#).

Examples

```
counts <- strat_simulate(c(.20, .35, .50), d = .08, association = .40,
                        n_bilateral = 200, n_unilateral = 50, seed = 12)
fit <- strat_fit(counts, model = "yule")
fit
strat_test(fit, d0 = 0, method = "LR")
strat_confint(fit, method = "Wald")
```

otitis_media	<i>Published Age-Stratified Otitis Media Treatment Counts</i>
--------------	---

Description

Aggregated bilateral and unilateral response counts from an acute otitis media trial comparing amoxicillin and cefaclor, reproduced in the analysis of Liang et al. (2024). The table contains 203 evaluable children in three age strata. It is published clinical data, not synthetic subject records.

Usage

```
data(otitis_media)
```

Format

A data frame with six rows and seven columns:

stratum Age stratum, labeled "age < 2", "age 2-5", or "age >= 6", in years.

group Treatment label, "Amoxicillin" or "Cefaclor". Amoxicillin precedes cefaclor within each stratum.

m0 Bilateral children with neither affected ear cured at day 14.

m1 Bilateral children with exactly one affected ear cured at day 14.

m2 Bilateral children with both affected ears cured at day 14.

u0 Unilateral children whose affected ear was not cured at day 14.

u1 Unilateral children whose affected ear was cured at day 14.

Details

A positive response means that an affected ear was effusion-free (cured) at the 14-day assessment. Bilateral and unilateral designation refers to the number of ears affected at baseline. A unilateral child contributes the outcome of the affected ear. These categories do not represent individual records reconstructed by treating an unobserved ear as missing.

The reproduced table contains 75 bilateral and 128 unilateral children, for 203 independent subjects and 278 affected-ear outcomes. There are 97 children in the amoxicillin group and 106 in the cefaclor group. The original trial enrolled 214 children with 293 affected ears; those enrollment totals differ from the evaluable counts in this table. The data object contains only the six published aggregate rows.

Use `reference = "Amoxicillin"` for a common marginal risk difference defined as cefaclor minus amoxicillin. A positive difference therefore means a higher probability of an effusion-free ear under cefaclor. The `m1` category counts children with one cured ear, not individual ears. Bilateral children count once in the subject total.

The direct transcription source is Table 26 of the version-2 preprint corresponding to Liang et al. (2024). Under the shared Dallal association model, its Table 28 reports a common risk-difference estimate of 0.1742. That estimate is useful as a numerical reproduction check and is distinct from the child-level proportion with all affected ears cured. The deterministic transcription script is included in the source repository as `'data-raw/otitis_media.R'`.

Source

Liang, S., Fang, K.-T., Huang, X.-W., Xin, Y. and Ma, C.-X. (2024). Homogeneity tests and interval estimations of risk differences for stratified bilateral and unilateral correlated data. *Statistical Papers*, **65**, 3499–3543. doi:10.1007/s00362024015326. Corresponding preprint, Table 26: <https://arxiv.org/abs/2304.00162v2>.

References

Mandel, E. M., Bluestone, C. D., Rockette, H. E., Blatter, M. M., Reisinger, K. S., Wucher, F. P. and Harper, J. (1982). Duration of effusion after antibiotic treatment for acute otitis media: comparison of cefaclor and amoxicillin. *Pediatric Infectious Disease*, **1**(5), 310–316. doi:10.1097/00006454-19820900000006. PubMed: <https://pubmed.ncbi.nlm.nih.gov/6760146/>.

See Also

[strat_data](#), [strat_fit](#), [yule_analyze](#)

Examples

```
data(otitis_media)
otitis_media

dat <- strat_data(otitis_media, reference = "Amoxicillin")
dat$N

# Published Dallal-model estimate, rounded to four decimal places: 0.1742.
fit <- strat_fit(dat, model = "dallal")
fit$d

# The same count data are accepted by all four association models.
models <- c("yule", "dallal", "rosner", "donner")
sapply(models, function(model) strat_fit(dat, model = model)$d)
```

paired_trial

Synthetic Paired Binary Outcomes for a Two-Group Study

Description

A reproducible teaching dataset with two groups, two strata, and both bilateral and unilateral observations. These are constructed illustrative data, not observations from a clinical study.

Usage

```
data(paired_trial)
```

Format

A data frame with 500 rows and five columns:

- id** Unique subject identifier.
- centre** Stratum label, "A" or "B".
- group** Group label, "control" or "treatment".
- left** Binary response coded 0 or 1.
- right** Binary response, or NA for a unilateral subject.

Details

Each group in each stratum has 100 bilateral and 25 unilateral subjects. The paired outcomes are ordered for illustration, with approximately half the discordant subjects having each orientation. This constructed dataset demonstrates input and analysis; it is not a simulation replicate or a basis for substantive conclusions. The source repository includes the deterministic construction script in 'data-raw/paired_trial.R'.

Source

Constructed for YuleInfer by Guanjie Lyu.

See Also

[yule_import](#), [yule_analyze](#)

Examples

```
data(paired_trial)
dat <- yule_import(paired_trial, group = "group",
  outcomes = c("left", "right"), stratum = "centre",
  reference = "control", id = "id")
yule_analyze(dat)
```

strat_confint

Confidence Sets under the Fitted Association Model

Description

Constructs LR or efficient-score confidence sets by test inversion, or a Wald interval, for the association family recorded in the fitted object. All detected components and unresolved regions are retained.

Usage

```
strat_confint(object, level = .95,
  method = c("LR", "score", "Wald"), control = list())
```

Arguments

object	An unconstrained, regular fit from strat_fit or yule_fit .
level	Confidence level strictly between zero and one.
method	"LR", "score", or "Wald".
control	A named list of inversion controls: • <code>grid.length</code>: initial grid size; default 33. • <code>refinements</code>: midpoint refinements; default 1. • <code>root.tol</code>: endpoint tolerance; default $1e-6$. • <code>boundary.eps</code>: endpoint padding; default $1e-5$. • <code>max.evaluations</code>: profile-fit budget; default 300. • <code>fit.control</code>: fitting overrides; the default empty list inherits the fitted object's controls.

See [yule_confint](#) for exact definitions and allowed ranges.

Details

LR and score confidence sets contain candidate $d \in (-1, 1)$ whose statistic does not exceed $qchisq(level, df = 1)$. Each numerical evaluation fits that family's constrained nuisance likelihood. The Wald interval uses the regular standard error and the normal critical value, intersected with the open physical range $(-1, 1)$.

Inversion uses grids, midpoint refinement, warm starts and endpoint root solving. The conservative LR tail bound applies to all four families. Yule score inversion uses an orthogonal-score bound. Dallal, Rosner and Donner score inversion use a separate bound based on counts incompatible with extreme margins. These conservative bounds exclude rejected tails of the regular stationary score test without ill-conditioned nuisance fits. Nonregular or unresolved profiles in the remaining search stay explicit. See [yule_confint](#) for the algorithm and both score bounds.

complete denotes success at the reported numerical resolution, not proof that no narrow components exist between grid nodes. Failed regions, unresolved tails or exhausted budgets produce complete = FALSE, with NA endpoints where appropriate. Increasing grid.length, refinements or max.evaluations can assess or improve resolution. Unknown regions are not silently joined into a single interval. Finite endpoints of a central component can coexist with complete = FALSE when remote profile regions remain unresolved; inspect the complete diagnostics before treating the global set as resolved.

Regular full fits are required. LR profiling can use a finite converged boundary nuisance fit; efficient-score inversion requires regular constrained fits. The chi-square calibration remains an interior large-sample approximation. For confint(fit), inherited S3 methods use this same model-aware inference engine.

Value

An object of class c("strat_ci", "yule_ci"), containing the following fields:

intervals A matrix with lower and upper endpoints, one row per detected component.

method, level The inference method and confidence level.

estimate The fitted common risk difference.

connected, complete Set topology and numerical completeness at the reported resolution.

diagnostics Search, profile and endpoint diagnostics; see [yule_confint](#) for definitions.

Data-frame conversion extracts components and their completeness status. Printing also retains this status.

See Also

[strat_fit](#), [strat_test](#), [yule_confint](#)

Examples

```
counts <- strat_simulate(c(.20, .35), .08, 1.30, 200, 50,
                        seed = 12, model = "rosner")
fit <- strat_fit(counts, model = "rosner")
interval <- strat_confint(fit, method = "Wald")
interval
as.data.frame(interval)
```

strat_data

Prepare Counts for Stratified Binary Data Inference

Description

Validates aggregated bilateral and unilateral subject counts and records treatment order. Data preparation does not select an association model.

Usage

```
strat_data(data, reference = NULL)
```

Arguments

data	A nonempty data frame with group and bilateral counts <code>m0</code> , <code>m1</code> , <code>m2</code> , unilateral counts <code>u0</code> , <code>u1</code> , or both complete sets. An omitted stratum column means one stratum.
reference	The reference-group label. If <code>NULL</code> , the first observed group is the reference.

Details

`m0`, `m1`, and `m2` count bilateral subjects with zero, one, or two positive responses. `m1` counts subjects, not organs. `u0` and `u1` count unilateral negative and positive subjects. Count columns must be numeric vectors of finite nonnegative integers, with one value per row. Lists, matrices, factors and text are not count vectors. Group and stratum labels must be vectors with no missing, empty or whitespace-only values; numeric labels must be finite. Column names must be nonmissing and unique. Exactly two groups and one row per stratum/group combination are required. Every combination must contain at least one subject. Omitting both unilateral columns means zero unilateral subjects. Omitting all bilateral columns means zero bilateral subjects. At least one complete set of counts is required. Omitting stratum creates a single stratum labelled "1". Partial count sets and missing values are rejected.

Strata retain their order of first appearance. Rows are normalized to the reference group followed by the second group within each stratum. The common risk difference is the second group's margin minus the reference group's margin. Specify reference to fix that direction.

Bilateral subjects count once toward the total subject count. Unilateral-only strata are supported; their unidentifiable associations are omitted in fitting. This constructor uses the same count checks as `yule_data`. Use `yule_import` for subject-level observations and `as.data.frame` to export a prepared object's normalized count table.

Value

A list of class `c("strat_data", "yule_data")`, with normalized counts, ordered strata and groups, reference, number of strata `J`, and total subjects `N`. The count table includes `n_bilateral` and `n_unilateral`.

See Also

[strat_fit](#), [strat_simulate](#), [yule_data](#)

Examples

```
counts <- strat_simulate(.30, .05, .40, 100, 25, seed = 12)
strat_data(counts, reference = "1")
```

strat_fit

Fit a Common Risk Difference under Four Association Models

Description

Maximum likelihood estimation for stratified bilateral and unilateral binary data, with a common marginal risk difference and stratum-specific native association shared between the two treatment groups.

Usage

```
strat_fit(data, d = NULL, control = list(),
          model = c("yule", "dallal", "rosner", "donner"))
```

Arguments

data	An object from strat_data or yule_data , or a count data frame accepted by those constructors.
d	NULL to estimate the common risk difference, or a finite value strictly between -1 and 1 to fix it and estimate nuisance parameters. The direction is group 2 minus the reference group.
control	Named fitting controls. Iteration and likelihood defaults are <code>maxit = 1000L</code> and <code>reltol = 1e-10</code> . Diagnostic tolerances default to <code>gradient_tol = 1e-5</code> and <code>boundary_tol = 1e-7</code> . Alternative starts default to <code>multistart = TRUE</code> . See the details and yule_fit . Unknown controls are rejected.
model	Association restriction; default "yule".

Details

For stratum j , reference risk b_j and treatment risk $b_j + d$ share one selected native association parameter. See [strat_models](#) for all definitions and feasible domains. The likelihood combines bilateral multinomial counts with unilateral binomial counts. The two observation types share the same marginal risk within each group and stratum.

Fitting uses analytic gradients and feasible transformed coordinates. Deterministic multistart checks and likelihood polishing provide safeguards. `maxit` is a positive integer limiting each optimizer call. `reltol` controls relative likelihood tolerance; `gradient_tol` controls natural-score stationarity after scaling by the square roots of information diagonals; `boundary_tol` controls proximity to marginal and bilateral-cell probability boundaries, with the additional Yule coefficient boundary check for Yule fits. Each tolerance must be strictly between zero and one. `multistart` selects alternative deterministic starts. Numerical safeguards do not certify a global maximum.

Associations in unilateral-only strata are unidentifiable and omitted from the fitted coordinates; their reported associations and bilateral probabilities are NA. Their unilateral likelihood contributions remain in marginal-risk estimation.

Expected information is returned in natural coordinates: `d`, all `baseline[stratum]` parameters, then identifiable `association[stratum]` parameters. Efficient information for `d` eliminates all nuisance coordinates. Yule marginal and association scores are orthogonal; the classical native coordinates generally require the full nuisance adjustment.

Inspect `convergence$regular` before inference. A numerically usable boundary fit may have `convergence$ok = TRUE` without being regular. Nonregular fits have `se = NA`; regular inferential calls reject them.

Value

An object of class `c("strat_fit", "yule_fit")`, with principal fields:

`model, data` The selected family and normalized counts.
`d` The common risk-difference estimate.
`baseline` Reference-group risks, one per stratum.
`association` Shared stratum estimates on the model's native scale.
`coefficients, probabilities` Natural parameters and fitted response probabilities.

logLik The complete sampling log likelihood.

score, information Natural scores and expected information.

efficient_information, se Nuisance-adjusted information and the regular standard error of the common risk difference.

convergence Numerical and regularity diagnostics.

fixed, controls Whether the risk difference was fixed, and the resolved fitting controls.

See [yule_fit](#) for shared diagnostic fields. Extraction methods are documented in [yule_methods](#).

Practitioner workflow

The getting-started vignette demonstrates preparation and reporting. The models-and-inference vignette compares four association restrictions.

```
vignette("getting-started", package = "YuleInfer")
vignette("models-and-inference", package = "YuleInfer")
```

See Also

[strat_models](#), [strat_test](#), [strat_confint](#), [strat_data](#), [yule_fit](#)

Examples

```
counts <- strat_simulate(c(.20, .35), .08, .70, 200, 50,
                        seed = 12, model = "dallal")
fit <- strat_fit(counts, model = "dallal")
summary(fit)
coef(fit)
fit$convergence$regular
strat_fit(counts, d = .08, model = "dallal")
```

strat_models

Association Models and Their Feasible Domains

Description

Definitions and parameter domains for the shared Yule, Dallal, Rosner and Donner association restrictions used by YuleInfer.

Details

For an exchangeable bilateral pair, let π be either response's marginal probability and q the joint-positive probability. Counts of zero, one and two positive responses have probabilities

$$p_0 = 1 - 2\pi + q, \quad p_1 = 2(\pi - q), \quad p_2 = q.$$

Strict positivity requires $0 < \pi < 1$ and $\max(0, 2\pi - 1) < q < \pi$. Exact boundaries are excluded.

The `model` argument selects the native association coordinate:

"yule" Yule's colligation coefficient (Yule's Υ),

$$\Upsilon = \frac{\sqrt{q(1-2\pi+q)} - (\pi - q)}{\sqrt{q(1-2\pi+q)} + (\pi - q)}.$$

Its domain is $-1 < \Upsilon < 1$ at every interior margin; independence is $\Upsilon = 0$. This is not Yule's Q .

"dallal" Conditional response probability $\gamma = P(Y_2 = 1 \mid Y_1 = 1) = q/\pi$, giving $q = \gamma\pi$. Its domain is $\max(0, 2 - 1/\pi) < \gamma < 1$. Some presentations use the complementary conditional probability; this package uses γ .

"rosner" Conditional-to-marginal risk ratio $R = P(Y_2 = 1 \mid Y_1 = 1)/\pi = q/\pi^2$, giving $q = R\pi^2$. Its domain is $\max(0, (2\pi - 1)/\pi^2) < R < 1/\pi$. Independence is $R = 1$.

"donner" Pearson correlation $\rho = (q - \pi^2)/\{\pi(1 - \pi)\}$, giving $q = \pi^2 + \rho\pi(1 - \pi)$. Its domain is $-\min\{\pi/(1 - \pi), (1 - \pi)/\pi\} < \rho < 1$. Negative correlations are supported when feasible. Every $0 \leq \rho < 1$ is feasible at every interior margin, and independence is $\rho = 0$.

For Yule's model, the paired-response odds ratio is $\Omega = \{(1 + \Upsilon)/(1 - \Upsilon)\}^2$. It describes association between units within a subject and is distinct from a marginal treatment odds ratio comparing the two groups.

In a fitted stratum, one native association parameter is shared between the two treatment groups. Its value must satisfy the bounds at both margins. Parameters may differ across strata. Although each coordinate can represent an individual positive exchangeable table, sharing different coordinates across unequal treatment margins imposes different likelihood restrictions. Feasibility does not establish correct specification.

The three classical options adapt these association restrictions to a stratified common-risk-difference likelihood with both bilateral and unilateral subjects. They are not exact reproductions of every test published under the classical names. In particular, Donner-model LR, score and Wald inference here differs from Donner's adjusted chi-square procedure implemented by `aod::donner()`.

Worked comparison

The registered models-and-inference vignette illustrates estimation, tests and confidence sets under the different restrictions.

```
vignette("models-and-inference", package = "YuleInfer")
```

References

- Dallal, G. E. (1988). Paired Bernoulli trials. *Biometrics*, 44, 253–257. <https://doi.org/10.2307/2531913>
- Rosner, B. (1982). Statistical methods in ophthalmology: An adjustment for the intraclass correlation between eyes. *Biometrics*, 38, 105–114. <https://doi.org/10.2307/2530293>
- Donner, A. (1989). Statistical methods in ophthalmology: An adjusted chi-square approach. *Biometrics*, 45, 605–611. <https://doi.org/10.2307/2531501>

See Also

[strat_probabilities](#), [strat_fit](#), [strat_power](#)

Examples

```
# These four native coordinates give the same table at pi = .5.
strat_probabilities(.5, .4, model = "yule")
strat_probabilities(.5, .7, model = "dallal")
strat_probabilities(.5, 1.4, model = "rosner")
strat_probabilities(.5, .4, model = "donner")
```

strat_power	<i>Own-Model Analytic Local Power for a Common Risk Difference</i>
-------------	--

Description

Calculates first-order local power for LR, efficient-score and Wald tests under a selected, correctly specified association model. These three tests share the same first-order power within that model.

Usage

```
strat_power(baseline, association, n_bilateral, n_unilateral = 0,
            d0 = 0, h = NULL, delta = NULL, alpha = .05,
            model = c("yule", "dallal", "rosner", "donner"))
```

Arguments

baseline	Reference-group marginal risks at the null, one per stratum.
association	A scalar or stratum-specific native association shared between groups. See strat_models .
n_bilateral, n_unilateral	Subject counts: a scalar, a length-two group vector, or a matrix with one row per stratum and two columns.
d0	Null common risk difference. Null treatment risks equal baseline + d0.
h	Local effect sizes in $d_N = d_0 + h/\sqrt{N}$; mutually exclusive with delta.
delta	Finite departures $d_N - d_0$, converted to $h = \sqrt{N} \delta$; mutually exclusive with h.
alpha	Two-sided significance level strictly between zero and one.
model	Correctly specified generating and testing family; default "yule".

Details

At an interior null with a fixed number of strata and fixed sampling proportions, expected information is evaluated in natural coordinates. Efficient information is the nuisance-adjusted Schur complement

$$I_{\text{eff}} = I_{dd} - I_{d\eta} I_{\eta\eta}^{-1} I_{\eta d}.$$

Yule marginal and association scores are orthogonal, yielding the simpler formula documented in [yule_power](#). Classical native coordinates generally require the full nuisance adjustment.

Noncentrality is $\lambda = h^2 I_{\text{eff}}/N$; power is the upper tail of $\chi_1^2(\lambda)$ beyond the central chi-square critical value. N counts subjects, not individual responses. Omitted h and delta give the null rejection probability.

The selected native association is held fixed along its own true local alternative path. Supplied null and finite alternative parameters must be feasible under that model. Positive and negative departures

have equal first-order power. These are local asymptotic approximations, not exact finite-sample powers.

This function does not calculate the power of one model's test along another model's true alternative path. Own-model information cannot be substituted for that cross-model comparison, even when the null tables coincide. No Pitman-efficiency interpretation is supplied for size-distorted, misspecified null tests.

Value

A data frame with N, d0, delta, h, total_information, information_per_subject, lambda, power, alpha, and model. The design attribute records the supplied null design, including its model.

References

Banerjee, M. (2005). Likelihood ratio tests under local alternatives in regular semiparametric models. *Statistica Sinica*, 15, 635–644. <https://www3.stat.sinica.edu.tw/statistica/j15n3/j15n33/j15n33.html>

See Also

[strat_models](#), [strat_fit](#), [strat_simulate](#), [yule_power](#)

Examples

```
strat_power(c(.20, .35), .70, 1000, 100,
            h = c(0, 1, 2), model = "dallal")
strat_power(c(.20, .35), 1.30, 1000, 100,
            delta = c(0, .01, .02), model = "rosner")
```

strat_probabilities	<i>Bilateral Probabilities under a Selected Association Model</i>
---------------------	---

Description

Computes the probabilities of zero, one and two positive responses in an exchangeable bilateral pair. Inputs are a marginal risk and a native association parameter.

Usage

```
strat_probabilities(pi, association,
                   model = c("yule", "dallal", "rosner", "donner"))
```

Arguments

pi	Finite marginal probabilities strictly between zero and one.
association	A finite native association parameter for the selected model: Yule's Y, Dallal's conditional probability, Rosner's conditional-to-marginal risk ratio, or Donner's Pearson correlation.
model	Association model. The default is "yule".

Details

pi and association must have equal lengths, or one must be a scalar. Scalar arguments are repeated. See [strat_models](#) for the definitions and margin-dependent feasible domains.

Each combination must yield strictly positive probabilities. Exact boundaries and infeasible combinations are rejected. Negative Donner correlations are accepted only within the margin-specific lower bound. Yule’s Y is supported throughout $(-1, 1)$ at any interior margin, subject to floating-point precision.

Value

A data frame with pi, association, p0, p1, p2, and q. The joint-positive probability q equals p2, and the marginal risk equals p2 + p1/2. association is on the selected model’s native scale. The model attribute records the selected family.

See Also

[strat_models](#), [strat_simulate](#), [strat_fit](#), [yule_probabilities](#)

Examples

```
strat_probabilities(c(.20, .35), .40, model = "yule")
strat_probabilities(c(.20, .35), .70, model = "dallal")
strat_probabilities(c(.20, .35), 1.30, model = "rosner")
strat_probabilities(.20, -.10, model = "donner")
```

strat_simulate	<i>Simulate Mixed Binary Counts under a Selected Association Model</i>
----------------	--

Description

Generates bilateral multinomial and unilateral binomial subject counts under a common risk difference and shared native association within each stratum.

Usage

```
strat_simulate(baseline, d, association, n_bilateral,
               n_unilateral = 0, seed = NULL,
               model = c("yule", "dallal", "rosner", "donner"))
```

Arguments

- baseline Reference-group marginal risks, one per stratum. Unique nonempty names are used as stratum labels when supplied.
- d Common risk difference: second group minus reference group.
- association A scalar native association or one value per stratum, shared between the two groups. See [strat_models](#).
- n_bilateral, n_unilateral Nonnegative integer subject counts: a scalar for all cells, a length-two group vector repeated across strata, or a matrix with one row per stratum and two columns.

seed	An optional nonnegative integer seed. Supplying a seed restores the previous global random-number state after simulation.
model	Generating association model; default "yule".

Details

Both baseline and baseline + d must lie strictly between zero and one. Each association parameter must be feasible at both groups' margins. Every group-stratum combination must contain subjects.

Subjects are independent, bilateral responses are exchangeable, and unilateral and bilateral subjects share the same per-unit marginal risk within each group and stratum. Without seed, the current R random state is used and advanced; with seed, its previous state, including the absence of a previous state, is restored.

Value

A count data frame accepted by [strat_data](#), with columns stratum, group, m0, m1, m2, u0, and u1. Group labels are "1" and "2".

See Also

[strat_probabilities](#), [strat_fit](#), [strat_power](#), [yule_simulate](#)

Examples

```
strat_simulate(c(.20, .35), .05, .70, 100, 25,
              seed = 42, model = "dallal")
strat_simulate(c(.20, .35), .05, 1.30, 100, 25,
              seed = 42, model = "rosner")
strat_simulate(c(.20, .35), .05, -.10, 100, 25,
              seed = 42, model = "donner")
```

strat_test	<i>Test a Common Risk Difference under the Fitted Association Model</i>
------------	---

Description

Runs a two-sided likelihood-ratio, efficient-score or Wald test using the association family recorded in the fitted object.

Usage

```
strat_test(object, d0 = 0, method = c("LR", "score", "Wald"))
```

Arguments

object	An unconstrained, regular fit from strat_fit or yule_fit .
d0	Null common risk difference, strictly between -1 and 1.
method	Case-sensitive statistic name: "LR", "score", or "Wald".

Details

The statistics are

$$T_L = 2\{\ell_p(\hat{d}) - \ell_p(d_0)\}, \quad T_S = U_d(d_0)^2/I_{\text{eff}}(d_0), \quad T_W = (\hat{d} - d_0)^2 I_{\text{eff}}(\hat{d}).$$

Efficient information eliminates the selected model's nuisance parameters. All three statistics use a one-degree-of-freedom chi-square reference. The calibration requires a correctly specified interior model, identifiable nuisance coordinates and nonsingular information in a large-sample setting with a fixed number of strata.

The full fit must be regular and unconstrained. LR and score tests also require a regular converged constrained fit, using the full fit's controls. Unavailable fits cause informative errors rather than reported p-values. Repeated LR/score calls can reuse a successful regular constrained fit when the null, family, data, estimates, controls and diagnostics agree.

No model argument is needed: it is recorded by the fit. Old objects without that field are treated as Yule fits. These tests adapt classical association restrictions to the common-RD likelihood; see [strat_models](#) for their scope.

Value

An `htest` object containing the statistic, $df = 1$, p-value, estimated and null risk differences, and convergence diagnostics for the full and, when fitted, constrained model.

See Also

[strat_fit](#), [strat_confint](#), [strat_models](#), [yule_test](#)

Examples

```
counts <- strat_simulate(c(.20, .35), .08, .40, 200, 50,
                        seed = 12, model = "donner")
fit <- strat_fit(counts, model = "donner")
strat_test(fit, d0 = 0, method = "LR")
strat_test(fit, d0 = 0, method = "score")
```

yule_analyze

Analyze a Common Risk Difference in One Call

Description

Fits a common marginal risk difference, tests one null value, and constructs the corresponding confidence set. A compact print method reports the signed risk difference and its equivalent in percentage points, together with the group direction and inference availability.

Usage

```
yule_analyze(data, reference = NULL,
             model = c("yule", "dallal", "rosner", "donner"),
             d0 = 0, method = c("LR", "score", "Wald"),
             conf.level = 0.95, control = list(), ci_control = list())

## S3 method for class 'yule_analysis'
```

```
print(x,
      digits = max(3L, getOption("digits") - 3L), ...)

## S3 method for class 'yule_analysis'
as.data.frame(x, row.names = NULL,
              optional = FALSE, ...)
```

Arguments

<code>data</code>	A count data frame accepted by strat_data , or a <code>strat_data/yule_data</code> object. Rows identify a stratum and group; <code>m0</code> , <code>m1</code> , and <code>m2</code> count bilateral subjects with zero, one, and two positive responses. <code>u0</code> and <code>u1</code> count unilateral negative and positive subjects. Supply a complete set of bilateral count columns, a complete set of unilateral count columns, or both. Omit stratum for one stratum. For one row per subject, call yule_import first.
<code>reference</code>	One observed group label. With <code>NULL</code> , an existing data object's reference is retained; a raw data frame uses its first observed group. The estimated risk difference is the other group minus this reference.
<code>model</code>	The native association restriction shared by the two groups within each stratum. See strat_models .
<code>d0</code>	Null common risk difference, a finite number strictly between -1 and 1 . The test is two-sided.
<code>method</code>	One of "LR", "score", or "Wald"; the same method supplies the test and confidence set.
<code>conf.level</code>	Confidence level strictly between zero and one.
<code>control</code>	Named fitting controls for strat_fit .
<code>ci_control</code>	Named confidence-set controls for strat_confint . Its <code>fit.control</code> field overrides selected fitting controls during confidence-set inversion. See that function's help.
<code>x</code>	A <code>yule_analysis</code> object.
<code>digits</code>	Number of significant digits used for printing.
<code>row.names</code>	Optional row name for the one-row data frame.
<code>optional</code>	A standard <code>as.data.frame</code> argument; the fixed result column names are preserved.
<code>...</code>	Reserved for compatibility with the S3 generics.

Details

This convenience function calls [strat_fit](#), [strat_test](#), and [strat_confint](#). It uses the same likelihood, numerical settings, and regularity rules as separate calls. The default model is Yule and the default inference method is LR.

Input data, model, null value, confidence level, and both sets of controls are validated before fitting; invalid arguments raise an error. Subsequent numerical failures are retained in diagnostics. A nonregular fit is retained for inspection, with unavailable inference represented explicitly. Warnings from the underlying functions are recorded and retain their usual R warning behavior. Test and confidence-set availability are assessed separately: a test can be unavailable even when confidence-set inversion returns a complete result.

The risk difference is an absolute change in probability, preserving its sign. For example, $d = 0.08$ corresponds to eight percentage points; it is not an eight-percent relative change.

Confidence sets can be disconnected or incomplete. The original object is always retained in `confint`. The data-frame method returns exactly one row, even when inference is unavailable. Its lower and upper columns are populated only for a complete, finite set with one component. A disconnected set is never replaced by its enclosing interval. The `intervals` list column retains the original component matrix, including unresolved endpoints in incomplete sets. `width` is the sum of component lengths for a complete, finite set, and NA otherwise. Numerical completeness is assessed at the resolution used by the underlying confidence-set procedure; inspect its diagnostics when needed. The print method also displays confidence-set diagnostic messages for complete sets, including notices about nuisance-boundary profiles. A complete set can still carry numerical information that should be reviewed.

Value

A `yule_analysis` list containing:

<code>fit</code>	The fitted object, or NULL when numerical fitting fails.
<code>test</code>	The <code>htest</code> object, or NULL when testing is unavailable.
<code>confint</code>	The original confidence-set object, including all components and numerical diagnostics, or NULL when no set can be constructed.
<code>model, method, d0, conf.level</code>	The requested analysis settings.
<code>reference, comparison, direction</code>	The group labels and risk-difference direction.
<code>n_subjects, n_strata</code>	Subject and stratum counts. A bilateral subject counts once.
<code>controls</code>	The supplied fitting and confidence-set controls.
<code>diagnostics</code>	Captured error messages and warnings for fitting, testing, and interval construction, plus the full fitting convergence diagnostics.
<code>call</code>	The original function call.

`as.data.frame` returns a one-row data frame with the signed estimate, percentage points, standard error, test statistic and p-value, confidence level and endpoints, inference availability and completeness indicators, component count, fit status, diagnostic messages, and an `intervals` list column. Unavailable numerical results are NA, never zero. The print method returns its input invisibly.

See Also

[strat_data](#), [strat_fit](#), [strat_test](#), [strat_confint](#), [strat_models](#)

Examples

```
counts <- data.frame(
  stratum = rep(c("A", "B"), each = 2),
  group = rep(c("control", "treatment"), 2),
  m0 = c(70, 60, 55, 48), m1 = c(20, 25, 28, 30),
  m2 = c(10, 15, 17, 22), u0 = c(20, 18, 16, 14),
  u1 = c(5, 7, 9, 11)
)
analysis <- yule_analyze(counts, reference = "control")
analysis
as.data.frame(analysis)[c("estimate", "p.value", "lower", "upper",
  "test_available", "ci_complete")]
```

```
# Inspect the original confidence set and its numerical diagnostics.
analysis$confint
```

yule_confint	<i>Confidence Sets for the Common Risk Difference</i>
--------------	---

Description

Construct likelihood-ratio or score confidence sets by numerical test inversion, or a Wald interval, under the association model recorded by the fit. Detected disconnected components and numerical uncertainty are retained. This compatibility entry point is model-aware; `strat_confint` is recommended for analyses using multiple association families.

Usage

```
yule_confint(object, level = 0.95,
              method = c("LR", "score", "Wald"), control = list())

## S3 method for class 'yule_ci'
print(x, digits = max(3L, getOption("digits") - 3L), ...)

## S3 method for class 'yule_ci'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)
```

Arguments

object	An unconstrained, regular yule_fit or strat_fit object. Old fits without a model field are treated as Yule.
level	Confidence level strictly between zero and one.
method	"LR", "score", or "Wald".
control	A named list of numerical controls, described below.
x	A yule_ci object.
digits	Number of significant digits used for printing.
row.names	Optional row names passed to as.data.frame.
optional	Passed to as.data.frame.
...	Additional arguments passed to matrix printing or data-frame conversion.

Details

LR and score confidence sets contain candidate values $d \in (-1, 1)$ for which the corresponding statistic does not exceed $qchisq(level, df = 1)$. Each numerical evaluation maximizes its own constrained nuisance likelihood. The Wald interval is $\hat{d} \pm z_{(1+level)/2} \widehat{se}(\hat{d})$, intersected with the open physical range $(-1, 1)$.

Conservative analytic lower bounds on the statistics exclude distant tails before fitting. The remaining search uses a global grid, extra nodes near the estimate, uniform midpoint refinement, cached warm starts, and endpoint root solving. All detected accepted components are retained. Excluded grid values with status "excluded_bound" are statistic lower bounds, rather than fitted test statistics; accepted candidates and endpoint roots use constrained fits. An isolated nonboundary classical score-profile convergence failure is retried once from the original full fit, using the same fitting

controls and requiring a regular fit without a material likelihood decrease. Both attempts count toward `max.evaluations`. Any retries are recorded in `diagnostics$profile_retries`; remaining failures stay unresolved.

Up to four completed confidence sets are retained locally in the fitted object for repeated requests with identical data, fit values, method, level and numerical controls. Changes to inputs used by profiling invalidate reuse. Incomplete sets are recomputed rather than memoized. Returned diagnostics describe the original computation of a reused set. The cache does not change the numerical search or the confidence-set definition.

Conservative statistic bounds exclude rejected extreme tails before numerical profiling. The LR bound applies to all four likelihood families. The Yule score bound uses its orthogonal marginal score; the classical score bound counts outcomes incompatible with extreme treatment margins. Score bounds apply only to regular, stationary constrained-score fits. They are not boundary-adjusted reference laws and are not substituted for fitted statistics at accepted candidates or roots. If the classical bound has no incompatible observations in a tail, it supplies no exclusion there. Unresolved regions not excluded by a valid bound remain explicit. The numerical-theory document retains the formulas, definitions and classical derivation. Locate it with:

```
system.file("doc", "score-tail-bounds.txt",
            package = "YuleInfer")
```

The complete flag records success at the reported numerical resolution: grid classifications and endpoint solves succeeded, the evaluation budget was respected, and component counts were stable at the last refinement when refinement was requested. It does not prove the absence of arbitrarily narrow components between nodes. Increase `grid.length` or `refinements` to assess resolution sensitivity. Failed regions produce `complete = FALSE`; adjoining unknown endpoints are NA, without silently bridging those regions. Connectedness is NA when numerical completeness is unresolved.

Every method requires a regular full fit. LR profiling may use a finite, converged boundary nuisance fit, but score inversion requires regular constrained fits. The chi-square reference remains an interior, large-sample calibration; see [yule_test](#).

Value

`yule_confint` returns a `yule_ci` list containing:

intervals A numeric matrix with columns lower and upper, one row per detected component. Unresolved endpoints are NA.

method, level, estimate The selected method, confidence level, and estimated common risk difference.

connected, complete Logical numerical diagnostics, with the qualifications given above.

diagnostics Search range, analytic tail bound, fitted or excluded grid values, root solves, failed fits, evaluation count, grid resolution, and controls.

For Wald intervals, the diagnostic field `physical_boundary_open` marks endpoints clipped to -1 or 1. These physical endpoints are excluded. The printing method returns `x` invisibly. Data-frame conversion returns the component endpoints with method, level, and completeness columns.

Numerical controls

grid.length Number of initial global nodes; default 33, allowed range 9 through 1001. Additional local nodes and refinement nodes are added.

refinements Number of uniform midpoint refinements; default 1, allowed range 0 through 8.

`root.tol` Absolute endpoint root tolerance; default 1e-6. Must lie strictly between zero and 0.01.

`boundary.eps` Search distance from physical endpoints -1 and 1 before analytic tail restriction; default 1e-5. Must lie strictly between zero and 0.1. Accepted components touching a search boundary remain unresolved.

`max.evaluations` Maximum number of constrained profile evaluations. The default is 300; values from 25 to 100000 are allowed. Analytic tail exclusions do not require a fit. Exhaustion is reported in the returned diagnostics.

`fit.control` Named fitting controls passed to constrained fits. The default empty list inherits `object$controls`; supplied entries override those controls. See [yule_fit](#).

Worked confidence sets

The `models-and-inference` vignette demonstrates complete and incomplete sets on clinical data.

```
vignette("models-and-inference", package = "YuleInfer")
```

See Also

[yule_fit](#), [yule_test](#), [strat_confint](#).

Examples

```
counts <- yule_simulate(c(.20, .35), d = .08, association = .4,
                      n_bilateral = 200, n_unilateral = 30, seed = 70)
fit <- yule_fit(counts)
interval <- yule_confint(fit, method = "Wald")
interval
as.data.frame(interval)

score_set <- yule_confint(fit, method = "score")
score_set$intervals
score_set$complete
```

yule_data

Validate Aggregated Bilateral and Unilateral Counts

Description

Checks subject counts and records the treatment ordering for a common marginal risk difference. Bilateral counts describe zero, one, or two positive responses per subject; unilateral counts describe zero or one.

Usage

```
yule_data(data, reference = NULL)
## S3 method for class 'yule_data'
print(x, ...)
## S3 method for class 'yule_data'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)
```

Arguments

<code>data</code>	A nonempty data frame with group and bilateral counts <code>m0</code> , <code>m1</code> , <code>m2</code> , unilateral counts <code>u0</code> , <code>u1</code> , or both complete sets. An omitted stratum column means one stratum.
<code>reference</code>	The label of the reference treatment group. If <code>NULL</code> , the first observed group is the reference.
<code>x</code>	A <code>yule_data</code> object.
<code>row.names, optional</code>	Passed to <code>as.data.frame</code> for the count table.
<code>...</code>	Arguments passed to printing or data-frame conversion.

Details

Each count column must be a numeric vector with one finite, nonnegative integer per row. Matrix-valued, list, factor and character count columns are rejected; recode counts explicitly before calling this function. Exactly two treatment groups and exactly one row per stratum/group combination are required. Every combination must contain at least one subject. Missing unilateral columns mean zero unilateral subjects. Omitting all three bilateral columns means zero bilateral subjects; at least one complete set of count columns must be present. An omitted stratum column is filled with "1". Missing values, negative or fractional counts, duplicate combinations, and incomplete strata are rejected.

Group and stratum columns must be vectors of nonmissing, nonempty labels; labels containing only whitespace are rejected. Numeric labels must be finite. Labels are converted to character, preserving leading and trailing whitespace in otherwise nonempty labels. Column names must be nonmissing and unique. The reference must be one observed label, not a list or matrix.

Strata retain their order of first appearance. Rows are reordered by stratum and then by reference group followed by the second group. The risk difference is always the second group's marginal probability minus the reference group's probability. Choose reference explicitly when input order should not determine the direction.

A bilateral subject contributes one subject to `N`, rather than two. Unilateral-only strata are allowed; their unidentifiable association parameters are omitted when fitting. [strat_data](#) provides the model-neutral constructor with the same validation and treatment-order rules. For one row per subject, use [yule_import](#) first.

Value

A `yule_data` list with normalized counts, ordered strata and groups, reference, number of strata `J`, and total number of subjects `N`. The count table includes `n_bilateral` and `n_unilateral`. `as.data.frame` returns the normalized count table, suitable for export.

See Also

[strat_data](#), [yule_fit](#), [yule_simulate](#)

Examples

```
counts <- data.frame(
  stratum = rep(c("A", "B"), each = 2),
  group = rep(c("control", "treatment"), 2),
  m0 = c(70, 60, 55, 48), m1 = c(20, 25, 28, 30),
  m2 = c(10, 15, 17, 22), u0 = c(20, 18, 16, 14),
```

```
u1 = c(5, 7, 9, 11))
yule_data(counts, reference = "control")
```

yule_fit

Fit a Stratified Common Risk Difference with Shared Yule Association

Description

Maximum likelihood fitting for mixed bilateral and unilateral binary data. The marginal risk difference is common across strata, and the Yule colligation coefficient is shared between groups within each stratum. This compatibility entry point fits the Yule model only. Use [strat_fit](#) to select among all four association models.

Usage

```
yule_fit(data, d = NULL, control = list())
```

Arguments

data	A yule_data object or a count data frame accepted by yule_data .
d	NULL to estimate the common risk difference, or a finite number strictly between -1 and 1 to fix it and maximize the nuisance likelihood. The direction is group 2 minus group 1 in the ordering recorded by yule_data.
control	A named list containing any of the fitting controls described below. Unrecognized or invalid controls produce an error.

Details

Let b_j denote the group 1 marginal risk, d the common risk difference, and v_j the stratum-specific Yule coefficient. The group 2 risk is $b_j + d$. The parameter domain is

$$-1 < d < 1, \quad \max(0, -d) < b_j < \min(1, 1 - d), \quad -1 < v_j < 1.$$

The likelihood combines multinomial counts for bilateral subjects with binomial counts for unilateral subjects. Both observation types have the same marginal risk within a group and stratum.

Fitting uses analytic gradients and smooth margin and association coordinates with BFGS optimization. By default, deterministic starts include empirical, neutral, positive, and negative association values, followed by likelihood polishing with safeguarded Fisher scoring. The scoring steps use analytic information blocks, preserve the parameter domain, and accept only likelihood-compatible improvements. BFGS remains a fallback for boundary or insufficiently stationary results. These starts are numerical safeguards, not a certificate of a global maximum. Internal profile fitting can use a previous fit as a warm start; warm starts are not a public argument of this function.

The supported controls and defaults are:

`maxit = 1000L` Maximum iterations for each optimizer call; a positive integer.

`reltol = 1e-10` Relative likelihood tolerance for initial BFGS optimization. Fallback BFGS polishing uses a tolerance no larger than $1e-13$; Fisher polishing checks score stationarity directly.

`gradient_tol = 1e-5` Stationarity tolerance for the maximum absolute natural score divided by $\sqrt{\max(I_{kk}, 1)}$. The fixed risk-difference coordinate is excluded when `d` is supplied.

`boundary_tol = 1e-7` Tolerance for proximity of a marginal probability, an observed-design bilateral cell probability, or $1 - |v_j|$ to zero.

`multistart = TRUE` Whether to use the alternative association starts.

Each numerical tolerance must be strictly between zero and one.

Strata with no bilateral subjects have no identifiable association parameter. Their unilateral terms remain in the likelihood, while their association coordinates are omitted. The reported association and bilateral probabilities in those strata are NA; no within-subject association is imputed.

The expected information is returned in natural coordinates: `d`, all `baseline[stratum]` parameters, then the identifiable `association[stratum]` parameters. Yule marginal and association scores are orthogonal. Consequently, efficient information for d is

$$I_{\text{eff}} = \sum_j \frac{A_{1j}A_{2j}}{A_{1j} + A_{2j}}, \quad A_{ij} = \frac{4n_{ij}}{\text{Var}(K_{ij})} + \frac{u_{ij}}{\pi_{ij}(1 - \pi_{ij})},$$

where K_{ij} is a bilateral subject's number of positive responses, and n_{ij} and u_{ij} count bilateral and unilateral subjects.

Inspect `convergence$regular` before asymptotic inference. `convergence$ok` denotes a numerically usable optimizer result; boundary fits can satisfy that criterion without being regular. A regular fit additionally requires interior estimates and nonsingular information. Nonregular fits have `se = NA`, and the inferential functions reject them. If no finite likelihood candidate is found, fitting produces an error.

Value

An object of class `yule_fit`, a list containing:

`data` Normalized count data and recorded group order.

`model` The fitted family, "yule".

`d` The fitted common risk difference.

`baseline` Named reference-group risks, one per stratum.

`association` Named stratum association estimates. Unidentified associations are NA.

`coefficients` The natural parameter vector, including the fixed `d` when supplied.

`probabilities` A data frame of fitted margins, associations, bilateral count probabilities, and the joint-positive probability.

`logLik`, `logLik_kernel`, `logLik_constant` The complete sampling log likelihood, its parameter-dependent kernel, and the multinomial and binomial combinatorial constant.

`score`, `information` Natural scores and expected Fisher information. A fixed fit retains the full interest–nuisance coordinate layout for constrained-score calculations.

`efficient_information`, `se` Efficient information for `d` and its regular asymptotic standard error. For a fixed fit, this scalar standard error describes the hypothetical free risk-difference coordinate; `vcov` instead treats the supplied `d` as known.

`convergence` Numerical and regularity diagnostics, described in the next section.

`parameter_count`, `fixed`, `controls` The number of estimated coordinates, whether `d` was fixed, and the resolved fitting controls.

Convergence diagnostics

The convergence list contains:

`ok, regular` Whether the selected fit is numerically usable and whether it satisfies the additional regularity conditions.

`status, reason` Optimizer status and a diagnostic message.

`gradient, scaled_gradient` Raw transformed gradient and standardized natural-score stationarity diagnostic.

`boundary` Whether boundary checks fail.

`starts` Number of distinct optimizer starting points.

`minimum_probability` Minimum bilateral cell probability over the observed sampling design; infinity when no bilateral cells exist.

`evaluations` Function and gradient evaluation counts for the last selected BFGS call. These are not totals across starts and polishing.

`likelihood_evaluations` Actual combined likelihood/gradient engine evaluations across the fit. Requests at the same point share one calculation.

`fisher_iterations` Number of accepted Fisher-scoring steps.

`fisher_backtracks` Number of scoring-step backtracking reductions.

`iterations` Missing because BFGS does not return an iteration count.

See Also

[yule_data](#), [yule_probabilities](#), [yule_test](#), [yule_confint](#), [yule_methods](#)

Examples

```
counts <- yule_simulate(
  baseline = c(.20, .50), d = .08, association = c(-.40, .50),
  n_bilateral = 300, n_unilateral = 100, seed = 2026
)
fit <- yule_fit(counts)
coef(fit)
fit$convergence$regular

fixed <- yule_fit(counts, d = .08)
c(free = as.numeric(logLik(fit)), fixed = as.numeric(logLik(fixed)))
```

yule_import

Import Binary Outcomes Recorded Once per Subject

Description

Converts individual-subject data in wide format to the validated bilateral and unilateral counts used by all association models in YuleInfer.

Usage

```
yule_import(data, group, outcomes, stratum = NULL, reference = NULL, id = NULL)
```

Arguments

<code>data</code>	A nonempty data frame with one row per independent subject.
<code>group</code>	A character string naming the treatment-group column. Exactly two observed groups are required. Unused factor levels do not count as groups.
<code>outcomes</code>	A character vector naming one or two distinct binary outcome columns, for example <code>c("left", "right")</code> . Each column must contain plain numeric 0/1 values or logical values, with NA for unobserved outcomes. Factors and character responses must be recoded explicitly before importing.
<code>stratum</code>	A character string naming a stratum column, or NULL to place all subjects in one stratum labeled "1". Both treatment groups must be represented in every stratum.
<code>reference</code>	The reference-group label. If NULL, the first group appearing in data is the reference. Set this explicitly to fix the direction of the risk difference.
<code>id</code>	An optional character string naming a subject-ID column. When supplied, IDs must be nonmissing, nonempty and unique across the entire data set, including across strata. Without this column, each row is assumed to represent a different subject.

Details

For numeric responses, 1 means positive and 0 means negative; for logical responses, TRUE means positive and FALSE means negative. The same coding must be used in both outcome columns.

A subject with two observed outcomes contributes to m_0 , m_1 , or m_2 , according to the number of positive outcomes. A subject with exactly one observed outcome contributes to u_0 or u_1 , regardless of which column is observed. With a single outcome column, all subjects are unilateral. A row with no observed outcomes is an error; the importer never silently drops rows or imputes outcomes.

Using one observed outcome as unilateral data describes the observation pattern; it does not adjust for informative missingness. The subject must be appropriately represented by the marginal response model used for the analysis. In particular, an unobserved outcome must not be coded as zero.

Group, stratum and ID labels may be character, factor, logical or numeric vectors and are converted to character labels. Labels must be nonmissing and cannot be empty or consist only of whitespace; numeric labels must be finite. Leading and trailing whitespace in nonempty labels is preserved. All selected columns must be distinct. Repeated subject IDs are an error even when they occur in different strata: construct globally unique IDs before importing if identifiers are reused for different subjects.

Strata retain their order of first appearance. The returned count table orders the reference group first within each stratum. The estimated risk difference is the other group's marginal response probability minus the reference group's probability. Reordering input rows can change the default reference; setting reference prevents this.

Value

A validated object of class `c("strat_data", "yule_data")`, as returned by `strat_data`. Its counts table contains stratum, group, m_0 , m_1 , m_2 , u_0 , u_1 , $n_{\text{bilateral}}$, and $n_{\text{unilateral}}$. Its total subject count N equals the number of input rows. No association model is selected during import.

See Also

[strat_data](#), [strat_fit](#), [yule_fit](#)

Examples

```

subjects <- data.frame(
  subject = paste0("p", 1:8),
  treatment = rep(c("control", "active"), each = 4),
  left = c(0, 1, 0, 1, 0, 1, 1, NA),
  right = c(0, 0, 1, NA, 0, 1, 0, 1)
)
counts <- yule_import(subjects, group = "treatment",
  outcomes = c("left", "right"),
  reference = "control", id = "subject")

counts$counts
counts$N

# A single observed outcome per subject needs no special count columns.
single <- data.frame(treatment = rep(c("control", "active"), each = 4),
  response = c(FALSE, FALSE, TRUE, TRUE,
    FALSE, TRUE, TRUE, TRUE))
yule_import(single, "treatment", "response", reference = "control")

# For multiple strata, supply the column name explicitly.
subjects$site <- rep(c("A", "B"), 4)
yule_import(subjects, "treatment", c("left", "right"),
  stratum = "site", reference = "control", id = "subject")

```

Description

Extract estimates, expected covariance, fitted marginal probabilities, sampling likelihood, subject counts, and confidence sets from fitted models fitted by [yule_fit](#) or [strat_fit](#). The inherited methods honor the association family recorded by the fit.

Usage

```

## S3 method for class 'yule_fit'
coef(object, ...)
## S3 method for class 'yule_fit'
vcov(object, ...)
## S3 method for class 'yule_fit'
logLik(object, ...)
## S3 method for class 'yule_fit'
nobs(object, ...)
## S3 method for class 'yule_fit'
fitted(object, ...)
## S3 method for class 'yule_fit'
summary(object, ...)
## S3 method for class 'yule_fit'
print(x, digits = max(3L, getOption("digits") - 3L), ...)
## S3 method for class 'summary.yule_fit'
print(x, digits = max(3L, getOption("digits") - 3L), ...)
## S3 method for class 'yule_fit'

```

```
confint(object, parm = "d", level = .95,
        method = c("LR", "score", "Wald"), ...)
```

Arguments

<code>object</code>	An object returned by <code>yule_fit</code> or <code>strat_fit</code> .
<code>x</code>	A <code>yule_fit</code> or <code>summary.yule_fit</code> object, as appropriate for the print method.
<code>digits</code>	Number of significant digits for printing.
<code>parm</code>	The common risk difference, specified by "d" or 1. Other parameters do not have confidence-set methods.
<code>level</code>	Confidence level strictly between zero and one.
<code>method</code>	Confidence-set method: likelihood ratio, efficient score, or Wald.
<code>...</code>	For <code>confint</code> , arguments forwarded to <code>yule_confint</code> , including its control list. For print methods, additional printing arguments. Unused by the other methods.

Details

`coef` returns the common risk difference, all group 1 stratum baselines, and identifiable native association parameters. Association coordinates are omitted for unilateral-only strata.

`vcov` uses the expected-information inverse in natural coordinates and requires a regular interior fit. The analytic block inverse uses the orthogonality of Yule marginal and association scores for Yule fits; classical native coordinates use the full information adjustment. When the risk difference was fixed, its covariance row and column are zero and the nuisance covariance is conditional on that known value.

`logLik` includes the multinomial and binomial combinatorial constants. Its `df` attribute counts only estimated coordinates; its `nobs` attribute counts subjects. A bilateral subject counts once, even though two binary responses are observed. `nobs` follows the same subject-count convention.

`fitted` returns the marginal risk for each group–stratum cell in the normalized data order. Joint probabilities are available separately in `object$probabilities`.

`summary` reports estimates and regular standard errors, with NA standard errors for nonregular fits. Printed summaries identify the family and native association coordinate. Coefficient-wise test statistics and p-values are not calculated automatically. Use `strat_test` for inference on the common risk difference.

`confint` delegates to `yule_confint` and returns its confidence-set object rather than the usual two-column coefficient matrix. It preserves all detected interval components and numerical completeness diagnostics. Inference requires an unconstrained regular fit.

Value

`coef, fitted` Named numeric vectors.

`vcov` A named covariance matrix.

`logLik` A likelihood object of class `logLik`.

`nobs` The total number of subjects.

`summary` A list of class `summary.yule_fit`.

`confint` A confidence-set object of class `yule_ci`.

Print methods The supplied object, returned invisibly.

See Also

[yule_fit](#), [yule_test](#), [yule_confint](#)

Examples

```
counts <- yule_simulate(.30, .05, -.30, 400, 100, seed = 42)
fit <- yule_fit(counts)
summary(fit)
fitted(fit)
nobs(fit)
logLik(fit)
vcov(fit)["d", "d"]
confint(fit, method = "Wald")
```

yule_power

Analytic Local Power for a Common Risk Difference

Description

Calculates first-order local power for the Yule likelihood-ratio, efficient-score, and Wald tests, which share the same asymptotic power. This compatibility entry point remains Yule-only. Use [strat_power](#) for own-model local power under all four families.

Usage

```
yule_power(baseline, association, n_bilateral, n_unilateral = 0,
           d0 = 0, h = NULL, delta = NULL, alpha = .05)
```

Arguments

baseline	Reference-group marginal risks at the null, one per stratum.
association	A scalar Yule coefficient or one coefficient per stratum.
n_bilateral, n_unilateral	Subject counts: a scalar, length-two group vector, or matrix with one row per stratum and two columns.
d0	The null common risk difference. Null treatment risks are baseline + d0.
h	A vector of local effect sizes in $d_N = d_0 + h/\sqrt{N}$. Mutually exclusive with delta.
delta	A vector of finite risk-difference departures from d0. Converted to $h = \sqrt{N} \delta$. Mutually exclusive with h.
alpha	Two-sided significance level strictly between zero and one.

Details

At an interior, correctly specified null with a fixed number of strata and fixed sampling proportions, define

$$A_{ij} = \frac{4n_{ij}}{\text{Var}(K_{ij})} + \frac{u_{ij}}{\pi_{ij}(1 - \pi_{ij})},$$

where K_{ij} is the bilateral count and $\text{Var}(K_{ij}) = 2\{\pi_{ij}(1 - \pi_{ij}) + q_{ij} - \pi_{ij}^2\}$. Yule marginal and association scores are orthogonal, giving

$$I_{\text{eff}} = \sum_j \frac{A_{1j}A_{2j}}{A_{1j} + A_{2j}}.$$

The noncentrality is $\lambda = h^2 I_{\text{eff}}/N$, and power is the upper tail of $\chi_1^2(\lambda)$ beyond the ordinary chi-square critical value.

These are local asymptotic approximations, not exact finite-sample powers. The association is held fixed along the true alternative path. Negative and positive departures have the same first-order power. The supplied finite-N alternative must have interior treatment margins. If both h and δ are omitted, null rejection probability is returned. N counts subjects, not individual organs.

The function calculates power for the proposed Yule model only. It does not assign Pitman efficiency to misspecified competitors or substitute their own-model information for power along a shared-Yule alternative.

Value

A data frame with N , d_0 , δ , h , total_information , $\text{information_per_subject}$, λ , power , and α . The design attribute records the supplied margins, association and counts.

References

Banerjee, M. (2005). Likelihood ratio tests under local alternatives in regular semiparametric models. *Statistica Sinica*, 15, 635–644. <https://www3.stat.sinica.edu.tw/statistica/j15n3/j15n33/j15n33.html>

See Also

[yule_fit](#), [yule_test](#), [yule_simulate](#)

Examples

```
yule_power(c(.03, .06, .10), .60, n_bilateral = 1000,
           n_unilateral = 100, delta = c(0, .005, .010))
yule_power(c(.20, .35), .40, n_bilateral = 200,
           n_unilateral = 50, d0 = .10, h = c(0, .5, 1))
```

yule_probabilities	<i>Bilateral Response Probabilities from Yule Association</i>
--------------------	---

Description

Returns the exchangeable bilateral probabilities for zero, one, and two positive responses at a given marginal probability and Yule coefficient. This compatibility entry point remains Yule-only; use [strat_probabilities](#) to select another native association model.

Usage

```
yule_probabilities(pi, association)
```

Arguments

pi	A finite numeric vector of marginal probabilities strictly between zero and one.
association	Yule's colligation coefficient, strictly between minus one and one. A finite numeric vector.

Details

Arguments must have the same length, or one must be a scalar. Scalar arguments are repeated. The inverse is evaluated using stable diagonal-cell calculations, including negative association and rare margins.

Yule's colligation coefficient is $(\sqrt{p_0 p_2} - p_1/2)/(\sqrt{p_0 p_2} + p_1/2)$. It is Yule's Y, not Yule's Q. Independence corresponds to zero. Every interior marginal probability admits every coefficient in $(-1, 1)$. Boundary probabilities are excluded; parameter combinations exceeding floating-point precision produce an error.

Value

A data frame with columns pi, association, p0, p1, p2, and q. The joint positive probability q equals p2; the marginal probability equals p2 + p1/2.

See Also

[yule_simulate](#), [yule_fit](#)

Examples

```
yule_probabilities(c(.10, .30, .70), association = .60)
yule_probabilities(pi = .20, association = c(-.50, 0, .50))
```

yule_simulate	<i>Simulate Stratified Mixed Binary Data under Shared Yule Association</i>
---------------	--

Description

Generates bilateral multinomial counts and unilateral binomial counts. The model has a common risk difference and shared Yule association. This entry point generates Yule data only. Other generating models are available through [strat_simulate](#).

Usage

```
yule_simulate(baseline, d, association, n_bilateral,
              n_unilateral = 0, seed = NULL)
```

Arguments

baseline	A numeric vector of reference-group marginal risks, one per stratum. Unique nonempty names are used as stratum labels when supplied.
d	The common risk difference: second group minus reference group.
association	A Yule coefficient shared between arms within each stratum. A scalar or one value per stratum.

n_bilateral, n_unilateral	Nonnegative integer subject counts. Each can be a scalar for all cells, a length-two vector of group counts repeated across strata, or a matrix with one row per stratum and two columns.
seed	An optional nonnegative integer seed. Supplying a seed restores the previous global random-number state after simulation.

Details

Margins baseline and baseline + d must lie strictly between zero and one, and Yule coefficients strictly between minus one and one. Every stratum/group must have a positive subject total. Bilateral subjects are exchangeable pairs and subjects are independent. Unilateral and bilateral subjects share the same per-unit marginal risk.

Value

A data frame accepted by [yule_data](#) with columns stratum, group, m0, m1, m2, u0, and u1. Group labels are "1" and "2".

See Also

[yule_probabilities](#), [yule_fit](#), [yule_power](#)

Examples

```
yule_simulate(c(.20, .35), d = .10, association = c(-.30, .60),
              n_bilateral = 100, n_unilateral = 25, seed = 42)
```

yule_test	<i>Test the Common Marginal Risk Difference</i>
-----------	---

Description

Likelihood-ratio, score, and Wald tests for a common risk difference in stratified bilateral and unilateral binary data. This compatibility entry point uses the association family recorded in the fitted object; [strat_test](#) is recommended for model-neutral analyses.

Usage

```
yule_test(object, d0 = 0, method = c("LR", "score", "Wald"))
```

Arguments

object	An unconstrained, regular yule_fit or strat_fit object.
d0	Null common risk difference, strictly between -1 and 1. The direction is the second treatment group minus the reference group.
method	Test statistic: "LR", "score", or "Wald". Names are case sensitive.

Details

For a full estimate $\hat{d}$, constrained log likelihood $\ell_p(d_0)$, efficient information I_{eff} , and constrained marginal score U_d , the statistics are

$$T_L = 2\{\ell_p(\hat{d}) - \ell_p(d_0)\}, \quad T_S = U_d(d_0)^2 / I_{\text{eff}}(d_0), \quad T_W = (\hat{d} - d_0)^2 I_{\text{eff}}(\hat{d}).$$

All three use an asymptotic chi-square distribution with one degree of freedom and a two-sided alternative. These calibrations require a correctly specified interior model, identifiable nuisance parameters, nonsingular information, and increasing sample sizes with fixed strata.

The full fit must be regular. LR and score tests also require a converged, regular constrained nuisance fit; their fitting controls are inherited from object. Boundary, singular, or unconverged fits cause an informative error rather than a reported asymptotic p-value. No confidence set is fitted automatically; use [yule_confint](#) separately.

Repeated LR and score calls can reuse up to four recent successful regular constrained fits associated with object. Changes to data, fitted parameters, fitting controls, likelihood values or convergence diagnostics invalidate that reuse. Failed or nonregular fits are not cached. The cache is local to the fitted object and does not alter test definitions. The selected association family is also checked before reuse. Old objects without a `model` field are interpreted as Yule fits.

Value

An object of class `htest`, containing the selected statistic, `df = 1`, p-value, estimated and null risk differences, and convergence diagnostics for the full and, when fitted, constrained model.

See Also

[yule_fit](#), [yule_confint](#), [yule_simulate](#)

Examples

```
counts <- yule_simulate(c(.20, .35), d = .08, association = .4,
                      n_bilateral = 200, n_unilateral = 30, seed = 70)
fit <- yule_fit(counts)
yule_test(fit, d0 = 0, method = "LR")
yule_test(fit, d0 = 0, method = "score")
```

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